Rex Ouyang

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EDUCATION

University of California, Berkeley **May. 2028**
B.A. Computer Science, B.A. Applied Mathematics; EECS Honors Program Berkeley, CA

- **Coursework:** Nonstochastic Statistics*, Stochastic Estimation and Control*, Applications of Stochastic Process Theory*, Random Processes in Systems*, Convex Optimization*, Topology and Analysis*, Deep Learning

* denotes graduate coursework

SELECTED PROJECTS & EXPERIENCE

Undergraduate Student Researcher, UC Berkeley EECS **May. 2026 – Present**
Researching sharp entropy bounds in Euclidean spaces and geometricity conditions under Prof. Thomas Courtade

- Studying information-theoretic functional inequalities using Convex Analysis and Stochastic Calculus
- Analyzing reductions from entropy optimization over probability measures to finite-dimensional covariances

Undergraduate Course Staff, UC Berkeley EECS **Incoming Aug. 2026**
1 out of 3 Selected Tutors for UC Berkeley EECS126: Probability and Random Processes, for Fall 2026 semester

- Will lead office hours, answer student questions, grade assignments and exams, and support administration

An Exposition of the Sharp Symmetrized Talagrand Inequality **May. 2026 – Present**
Research project studying a recent result on optimal transport bounds measuring distance to centered distributions

- Using Convex Analysis and Stochastic Calculus, provided alternate proofs to key results in this report
- Building Monte Carlo simulations on probability distributions to test and visualize the inequality numerically

Vectorized Entropy-Based Wordle Solver **Mar. 2025 – Jul. 2026**
Built an information-theoretic solver for NYT Wordle that treats each game as sequential Bayesian inference, choosing the guess that maximizes expected entropy gain over the posterior of remaining candidate words

- Designed a two-tier candidate model that falls back from the 2,315 curated answers to the full 12,972-word legal list, making the solver robust to out-of-vocabulary answers that break naive filtering
- Averaged 3.483 guesses per game across simulated play, outperforming the human average of 3.92 guesses

Simulations Developer, Formula Electric at Berkeley **Sep. 2024 – May. 2026**
Analyzed measured current-draw time-series to model battery discharge dynamics under varying load, deriving a reduced formulation that estimated battery state faster than the team's existing numerical simulation

- Collaborated within a multidisciplinary engineering team, contributing the model to the shared simulation codebase for integration by team lead

Sparse Attack Detection in Power Networks via Lasso Optimization **Nov. 2025 – Dec. 2025**
Formulated sensor attack detection as a sparse recovery problem as an L1-regularized regression problem over measurements corrupted by gaussian noise and sparse adversarial injections.

- Built a Monte Carlo experiment generating corrupted measurement datasets across 100 randomized trials
- Achieved exact support recovery, every attacked node and line with zero false positives, in 95% of trials

RISC-V Digit Classification **Jun. 2025 – Jul. 2025**
Built a neural network in RISC-V assembly to classify handwritten digits from the MNIST dataset using matrix multiplication and file I/O routines.

- Implemented low-level numerical computation and data processing pipelines directly in RISC-V assembly
- Applied matrix-based computation techniques to support inference on digit classification

TECHNICAL SKILLS

- **Quantitative Methods:** Probability, Statistics, Stochastic Processes, Stochastic Analysis, Convex Optimization, Sparse Recovery, Statistical Modeling, Information Theory, Dynamical Systems, Machine Learning
- **Programming / Systems:** Java, Python, C/C++, SQL, Go, RISC-V Assembly, x86 Assembly
- **Scientific Computing / Tools:** NumPy, SciPy, CVXPY, Matplotlib, Git, GDB, Valgrind, Logisim, Unity